

EE 105 Fall 2025

Homework 3 – upload to grade scope

(Due February 24, before class, late submission will incur 20% points/day penalty)

Instructions: Perform the following tasks based on the circuits and concepts discussed in class. Be sure to show all work where applicable.

Problem 1: What are the different types of materials used in electronics? What are their charge carrier densities?

Problem 2: Why is silicon used in electronics?

Problem 3: Draw the atomic models of the p-type and n-type semiconductors.

Problem 4:



- What is this electronic device?
 - Draw the I-V characteristics and explain what the threshold voltage is.
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Problem 5: Write the ideal diode equation and explain each quantity in the equation.

- A silicon diode is connected in a circuit with a 5V power supply and a series resistor of $1\text{k}\Omega$. Draw the circuit. The saturation current of the diode is $I_s = 10^{-12}\text{A}$. Use the diode equation to calculate the current through the diode when the voltage across it is 0.7V.

$$n = 1$$

$$\frac{kt}{q} = 26\text{mV}$$

- Find the voltage across the diode when the current through it is 1mA.
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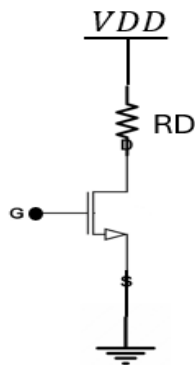
Problem 6: What is a MOSFET? What is a p-channel and n-channel MOSFET? Draw their side view and label each segment.

- A n-channel MOSFET is used in a circuit with a drain resistor $R_D=1\text{k}\Omega$ and a supply voltage $V_{DD}=10\text{V}$. The MOSFET has the following parameters:

$$V_{th}=2\text{V}$$

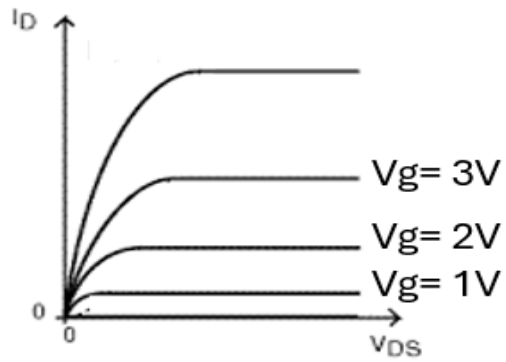
$$\frac{\mu_n \cdot C_{ox} \cdot Z}{2L} = 0.5 \frac{\text{mA}}{\text{V}^2}$$

If $V_{GS}=4\text{V}$, assuming the MOSFET is in the saturation region, calculate the current and voltage across the resistor.



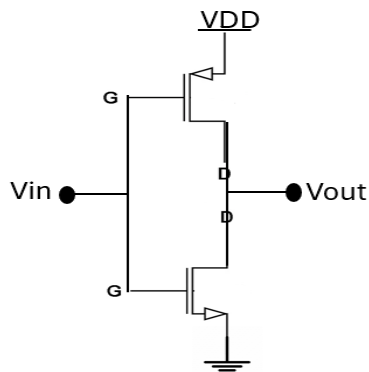
Problem 7: Draw the circuit symbol of p-channel and n-channel MOSFET.

Problem 8:



In the shown plot, draw another curve when $V_g < V_{th}$. What is the state of the N-channel MOSFET when $V_g < V_{th}$?

Problem 9: Fill up the table.



V_{in}	V_{out}
0	
VDD	

- What is this electronic device? Where can you use such a device?
